Question:

A mother brings her 3-year-old daughter to the pediatrician because she is concerned about her development. She states that her daughter seemed to regress in her motor development. Furthermore, she states she has been having brief episodes of uncontrollable shaking, which has been very distressing to the family. During the subsequent work-up, a muscle biopsy is obtained which demonstrates red ragged fibers and a presumptive diagnosis of a genetic disease is made. The mother states that she has another 6-year-old son who does not seem to be affected or have any similar symptoms. What genetic term explains this phenomenon?Answer Choices:A. Allelic heterogeneityB. Phenotypic heterogeneityC. Genetic heterogeneityD. Heteroplasmy

Reasoning + Answer:

<think> To answer this question, we need to consider the phenomenon of a genetic disease that affects one child but not another, despite both children having the same parents. One possible explanation for this phenomenon is allelic heterogeneity, which occurs when a genetic disease is caused by different alleles of a gene, even though the genes are the same. This is consistent with the information in the question, which states that the mother has another son who is not affected, suggesting that the genetic disease may be caused by a different allele in the affected child. The other options are not as likely to explain the phenomenon, as phenotypic heterogeneity refers to the different ways in which a single genetic disease can present, and genetic heterogeneity is a broader term that refers to the presence of different genetic mutations in a population. Heteroplasmy is a term that is typically used to describe the presence of different mitochondrial DNA in a cell, which is not relevant to this question. Therefore, based on the information provided, it is most likely that the genetic term that explains this phenomenon is allelic heterogeneity.</think><answer>Allelic heterogeneity</answer>